

The Subsector Composition of Portugal’s General-Government Balance, 1977–2025

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Abstract

A general-government balance is normally reported as a single number, although it is the sum of institutionally distinct subsectors whose balances are determined by different revenue bases, different expenditure mandates and different legal constraints. This paper asks how Portugal’s general-government balance has been composed across those subsectors over 49 years, and how persistent each subsector’s contribution has been.

We assemble a harmonised annual panel of net lending and net borrowing for General Government and its three subsectors from 1977 to 2025, keeping the **1995** statistical splice explicit rather than chaining across it, and decompose the aggregate balance exactly into its subsector components and their annual changes. The decomposition is an accounting identity, not a model: it reallocates a recorded quantity and estimates nothing.

Four results stand out. First, the composition is highly asymmetric and stable: the aggregate balance is positive in only 4 of 49 years (2019, 2023, 2024 and 2025), the Central Government balance is negative in all 49, and the Social Security balance is positive in 43. Second, in every year with a positive aggregate balance the combined non-Social-Security balance is negative and the Social Security balance exceeds it in absolute size; we quantify this as an offset ratio and decline to read it as a counterfactual. Third, magnitudes differ sharply across the 1995 splice — the aggregate balance averages -6.43% of GDP before it and -4.04% after — so no level statistic is pooled across the boundary. Fourth, the permanently negative Central Government headline balance does not imply a permanently negative underlying one: once interest is excluded, the Central Government primary balance is positive in 15 of the 45 years for which detailed accounts exist, spread across the panel rather than confined to recent years. We then separate two Social Security objects that are routinely conflated, the ESA 2010 Social Security Funds sector and the budget-execution systems published by the Portuguese Public Finance Council, and show that they differ in every year they overlap. No causal, normative or policy-intent interpretation is assigned to any result. Every figure and table is generated from a persisted artefact, and every number in this text is a macro read from one.

Keywords: general government; institutional subsectors; social security funds; accounting decomposition; fiscal statistics; Portugal

JEL: H61, H62, H55, E62

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1 Introduction

A country’s fiscal position is normally reported as one number. Portugal recorded a general-government balance of 2,059 million euro in 2025, or 0.67% of GDP, and it is that figure which enters fiscal rules, credit assessments and public debate.

The number is an aggregate of institutionally distinct parts. Under ESA 2010 the general government sector S.13 is the consolidation of central government (S.1311), state government (S.1312), local government (S.1313) and social security funds (S.1314) (Eurostat, 2013). These are not administrative subdivisions of one budget. They have different revenue bases — general taxation in one case, earmarked social contributions in another — different expenditure mandates, and different legal constraints on borrowing. Their balances are therefore determined by different things, and there is no reason to expect them to move together.

Aggregation hides that. In 2025 Portugal’s positive aggregate balance was the sum of a Central Government balance of -5,636 million euro, a Regional and Local balance of 630 million, and a Social Security Funds balance of 7,065 million. The headline figure and the composition behind it support quite different descriptions of the same year, and only the headline figure is routinely reported.

1.1 Research questions

This paper addresses two questions.

1. **How has Portugal’s general-government balance been composed across institutional subsectors since 1977, and how persistent are the contributions of Central Government, Regional and Local Government, and Social Security Funds?**
2. **What accounting mechanisms account for the changing contribution of the Social Security Funds, particularly in the recent period?**

Both are descriptive questions about recorded quantities, and we answer them as such. The central object is an accounting identity,

$$B_t^{GG} = B_t^C + B_t^{RL} + B_t^{SSF},$$

which holds by construction. Decomposing it reallocates an observed number; it estimates nothing and identifies nothing. We are explicit about this throughout, because the decomposition invites a counterfactual reading — that the aggregate balance would have been different under a different institutional arrangement — which the identity cannot support.

1.2 Related work and where this paper sits

Three strands of existing work bear on these questions, and each leaves a different part of them open.

The first is the statistical framework itself. ESA 2010 defines the subsectors and requires member states to report deficit and debt for each of them, and Eurostat and the ECB publish the resulting series (Eurostat, 2013, 2024b, 2025; European Central Bank, 2025). The composition of the aggregate balance is therefore not hidden: it is published. What the framework provides is a definition and a data supply, not an analysis of how a particular country’s composition has behaved over time.

The second is the literature on Portuguese fiscal sustainability, which works predominantly at the aggregate level and over long horizons. [Marinheiro \(2006\)](#) tests the sustainability of Portuguese fiscal policy on more than a century of annual data, using stationarity and cointegration tests on aggregate revenue, expenditure and debt. That question — whether the aggregate path satisfies

a solvency condition — is a different question from ours, and it is answered without decomposing the aggregate.

The third concerns Portuguese Social Security financing specifically. [Cardoso \(2019\)](#) examines the equity and sustainability of the system’s financing, noting that social-protection entitlements follow a dynamic that at times diverges from the cyclical path of the revenue that funds them; the Portuguese Public Finance Council has also examined the Social Security Financial Stabilisation Fund in the same context ([Portuguese Public Finance Council, 2024](#)). This literature is concerned with the system’s own finances and its long-run adequacy, rather than with the arithmetic by which that system’s balance enters the national aggregate.

A fourth strand is relevant less as a precedent than as a caution. Work on stock-flow adjustments has shown that the reported balance is an incomplete account of what happens to debt: [von Hagen and Wolff \(2006\)](#) find that debt accumulates by more than accumulated deficits and read the residual as evidence of accounting responses to fiscal rules, while [Seiferling \(2013\)](#) argues on integrated public-finance data that the residuals are smaller than previously assumed and are not well explained by fiscal transparency. We take no position in that debate. We take from it the discipline that a balance is a recorded accounting quantity whose relation to anything else has to be demonstrated rather than assumed — which is why this paper stops at the accounting and says so.

To our knowledge no study assembles and analyses the subsector composition of the Portuguese general-government balance systematically over a window of this length. We state that as a gap in the work we have found rather than as a claim of priority: the underlying series are published, and any competent user of the Eurostat or CFP data could construct parts of what follows. The contribution is in the assembly, the harmonisation, the discipline applied to the resulting statistics and the reproducibility of the whole, not in access to anything unavailable.

1.3 Contribution

We make six contributions.

1. **A harmonised long-run panel.** An annual panel of net lending and net borrowing for General Government and its three subsectors covering 1977–2025, built by splicing the Banco de Portugal / INE historical long series to the modern national accounts. The **1995** methodological boundary is retained as a documented splice, both vintages of the overlapping year are kept, and nothing is chained, smoothed or calibrated across it.
2. **An exact decomposition, in levels and in changes.** The aggregate balance and each of its annual changes are decomposed into subsector components that close to the sources’ rounding. Changes are additionally reported scaled by current-year GDP, which makes episodes forty years apart comparable without turning the decomposition into a ratio that no longer adds up.
3. **Persistence stated per statistical regime.** Sign frequencies, run lengths and one-year sign transitions, with magnitudes computed inside each regime rather than pooled across the splice. Pooling is not a presentational choice here: the aggregate balance averages -6.43% of GDP before 1995 and -4.04% after, and the pooled -4.91% describes neither period.
4. **Attribution of the largest episodes to revenue and expenditure.** The largest annual improvements and deteriorations, each attributed to a subsector and then to the revenue and expenditure movements that composed it.
5. **Two Social Security boundaries, separated and quantified.** The ESA 2010 Social Security Funds sector and the budget-execution systems published by the CFP are different accounting objects. We report them side by side, never add them, and measure the gap between them — which is non-zero in every one of the 7 years they overlap.

6. **A validation layer that separates two different tests.** Identity closure asks whether an extraction is arithmetically self-consistent; source agreement asks whether two independently published sources report the same number. The first cannot establish the second. Here the identities close to 1.00 million euro while two published sources still disagree by 66.8 million euro on one subsector-year.

1.4 Principal findings

The composition is asymmetric and it is stable. Over 49 years the aggregate balance is positive in 4 (2019, 2023, 2024 and 2025). The Central Government balance is negative in all 49 observations. The Social Security balance is positive in 43, with a longest unbroken run of 24 years. Regional and Local Government is the only subsector that changes sign with any regularity, and it is small in both directions.

In each of the 4 years with a positive aggregate balance, the combined non-Social-Security balance is negative and the Social Security balance exceeds it in absolute size. In 2025 the non-Social-Security balance was -5,006 million euro against a Social Security balance of 7,065 million, an offset ratio of 1.411. This is an arithmetic relation between two recorded numbers and we do not present it as anything else.

Magnitudes are regime-dependent and must be read that way. The Central Government balance averages -6.61% of GDP over 1977–1994 and -4.67% over 1995–2025; the Social Security balance rises from 0.46% to 0.72%.

The headline sign is not the underlying sign. Central Government records a negative balance in every year of the panel, but its primary balance — the balance excluding interest — is positive in 15 of the 45 years for which the detailed accounts exist, from 1986 to 2025. Interest averaged 4.17% of GDP over that panel and peaked at 7.73%. Both statements are descriptive and they are not in conflict; interest is the arithmetic that separates them. A conclusion that Central Government runs a permanent underlying deficit does not follow from the headline series.

Finally, the recent movement in the Social Security position is concentrated in one of the two internal budget systems. The Previdential balance moves from 2,333 million euro in 2019 to 6,712 million in 2025, while the Citizenship balance stays small in both directions and stands at -55 million. That is a description of the published series and not an account of what caused it.

1.5 Roadmap

Section 2 defines the institutional and accounting objects, and in particular why Central Government is not a synonym for the State in the budgetary sense. Section 3 describes the sources, the harmonisation, the 1995 splice, the vintages and the cross-source validation. Section 4 states the decompositions and the persistence measures. Section 5 presents the results. Section 6 sets out the accounting boundaries and the robustness of the weaker statistics. Section 7 concludes with what the evidence does and does not establish. Appendix A reports the change-point sensitivity in full and Appendix B the reproduction path.

2 Institutional and Accounting Framework

2.1 The subsectors

Under ESA 2010 the general government sector S.13 consolidates four subsectors ([Eurostat, 2013](#)):

- **S.1311, central government**, comprising all government units with a national sphere of competence *with the exception of social security units* ([Eurostat, 2024a](#));
- **S.1312, state government**, which does not apply to Portugal;

- **S.1313, local government**, which in the Portuguese case is reported together with the autonomous regions and appears here as Regional and Local Government;
- **S.1314, social security funds**, comprising all social security units *regardless of the level of government that operates or manages the scheme* (Eurostat, 2024b).

The two emphasised clauses matter for everything that follows, and they are the source of a recurring misreading.

Central Government in the national accounts is not the State in the budgetary sense. The statistical subsector excludes social security units by definition, and it includes many entities — public institutes, funds, reclassified public corporations — that no reader of a State Budget would think of as the State. A statement about S.1311 is therefore not a statement about the *Orçamento do Estado*, and the two should not be substituted for one another. Where this paper says Central Government it means S.1311 throughout.

The social security subsector is defined by function, not by tier. S.1314 collects social security units whichever level of government runs them. It is consequently not a level of administration sitting beneath Central Government; it is a functionally delimited sector alongside it.

2.2 The identity

The balancing item is B.9, net lending (+) or net borrowing (−), and it is the difference between total revenue and total expenditure for the sector concerned (Eurostat, 2025). Consolidation across subsectors means the aggregate is their sum:

$$B_t^{GG} = B_t^C + B_t^{RL} + B_t^{SSF}. \quad (1)$$

Equation (1) is definitional. In published data it closes only to the rounding of the published series, and we carry that residual as a column rather than assume it away; over the whole panel its largest absolute value is 1.00 million euro, which is the rounding unit of the sources.

Because the subsectors are consolidated, transfers between them net out of the aggregate but not out of the components. A transfer from Central Government to the Social Security Funds worsens B^C and improves B^{SSF} while leaving B^{GG} unchanged. This is central to interpreting the composition: the *location* of a balance within general government depends on the transfer convention, even though the aggregate does not. We return to this in Section 6 and deliberately do not construct a transfer-adjusted balance and call it underlying.

2.3 Two things the identity cannot do

It cannot support a counterfactual. Writing $B_t^{nonSSF} = B_t^C + B_t^{RL}$ and observing that $B_t^{SSF} > |B_t^{nonSSF}|$ in a given year says that two recorded numbers stand in a particular ratio. It does not say what the aggregate balance would have been had the Social Security Funds been organised differently, because the counterfactual would change contribution rates, entitlements and transfers simultaneously, and the identity contains no model of any of them.

It cannot support an attribution of responsibility. A subsector that accounts arithmetically for most of an annual improvement has not been shown to have produced it. The decomposition is silent on intent, on policy and on causation, and we use the word contribution in its arithmetic sense only.

3 Data and Harmonisation

3.1 Sources

The canonical balance panel takes 1977–1994 from the Banco de Portugal / INE historical long series (Banco de Portugal, 2023) and 1995–2025 from INE national accounts distributed through

PORDATA ([PORDATA, 2026](#)). Detailed revenue, expenditure, interest, investment, Maastricht debt and stock-flow data for the modern period come from the ESA 2010 annual workbooks published by the Portuguese Public Finance Council ([Portuguese Public Finance Council, 2026](#)), which also serve as an independent cross-check on the balance series. Table 1 records each file, the coverage taken from it, its vintage and a digest of the retained copy.

Every workbook is parsed programmatically from the bundled file. No value in this paper was transcribed by hand, including the values in this sentence: they are macros written by the pipeline from the persisted artefacts.

Table 1: Bundled sources, the coverage taken from each, the vintage of the retained file and the first twelve hexadecimal digits of its SHA-256 digest.

Source	Institution	Coverage used	Vintage	SHA-256 prefix
Banco Portugal Long Series	Banco de Portugal / INE	1977-1995 for the historical segment used here	2023-12	d3570ea90f78
PORDATA Balance By Level	INE via PORDATA	1995-2025	2026-04	2a5d54649041
CFP Annual General Government	Portuguese Public Finance Council (CFP)	1995-2025	2026-04-15	8f6b5a2916a4
CFP Annual Subsectors	Portuguese Public Finance Council (CFP)	2000-2025	2026-04-15	4d9e1b07cd38
CFP Social Security Detail	Portuguese Public Finance Council (CFP)	2019-2025 systems; detailed 2024-2025 tables	2026-04	d9f28e424ab6

3.2 Vintage

A fiscal series has a vintage as well as a coverage, and the two are different things. The modern national-accounts and CFP files used here are the April 2026 releases. Within them, 2024 and 2025 are flagged **provisional** by the publisher, and that flag is carried through the harmonisation into the canonical panel rather than dropped.

The distinction is not pedantry in this application. The provisional years are the years a reader is most likely to be interested in, and they are the years most likely to be restated. Any statement below about 2024 and 2025 should be read as a statement about the April 2026 vintage of those years.

3.3 Two statistical regimes, not one series

The splice at **1995** separates two source families built on different methodologies. We retain it explicitly. Nothing is chained, smoothed, interpolated or calibrated across it, and two consequences are enforced throughout the paper:

1. no statistic whose value depends on the *level* of the balance — a mean, a segment mean, a regression intercept — is computed across the boundary;
2. no annual change is computed through it, so the 1994-to-1995 change, which would mix a vintage revision with an economic movement, appears nowhere in the results.

1995 exists in both source families, so rather than choosing silently we keep both vintages and report the revision between them. The revisions are small in level terms but non-zero for every subsector, which is the direct evidence that the two regimes are not interchangeable at the level of a single year.

3.4 An explicit gap

Detailed revenue and expenditure components are available for General Government continuously from 1977. For the three subsectors they exist for 1977–1995 and 2000–2025 only. The 4 intervening years are absent from both source families and are therefore absent here: they are not imputed, no statistic is computed across them, and every figure drawn from the account panel breaks its line rather than joining 1995 to 2000.

The gap affects the detailed components only. The canonical B.9 panel has an observation for every one of the 49 years and for every subsector, which is why the composition results in

Section 5 cover the full window while the revenue–expenditure results do not. The detailed account panel holds 184 sector-year observations against the 4×49 a complete panel would have, and Table 2 states the two coverages separately: reporting a single observation count for a panel that has two would be misleading.

Table 2: Coverage of the detailed account panel. The canonical balance panel is complete; only the detailed components carry the 1996–1999 gap.

Subsector	First year	Last year	Observations
General Government	1977	2025	49
Central Government	1977	2025	45
Regional and Local	1977	2025	45
Social Security Funds	1977	2025	45

3.5 Identity closure is not source agreement

Two different validation questions are easy to conflate, and conflating them overstates what the checks establish.

An **identity** check asks whether an extraction is arithmetically self-consistent: whether B.9 equals the sum of its subsectors, whether revenue minus expenditure reproduces the recorded balance, whether the change in debt reconciles with the balance and the stock-flow adjustment. Such a check can close to numerical precision while both sources are wrong in the same way, because it never consults a second source.

A **source-agreement** check asks whether two independently published sources report the same number for the same year. Identity closure cannot establish it.

Table 3 reports both in one unit. The identities close to 1.00 million euro, the rounding of the published series. The largest disagreement between the PORDATA bridge and the CFP workbook is 66.8 million euro, on Central Government in 2002. That gap is nearly two orders of magnitude larger than any identity residual, which is the concrete reason for reporting the two separately: a paper that reported only the identity residuals would imply a precision the underlying statistics do not have.

Table 3: Identity closure and source agreement, in one unit. The two answer different questions and neither substitutes for the other.

Check	Quantity compared	N	Largest absolute difference (M EUR)	Where
Accounting identity	Aggregate balance minus the sum of its three subsectors	49	1.000	2004
Accounting identity	Revenue minus expenditure minus the recorded balance	184	0.000	Central Government, 1994
Accounting identity	Debt change against minus the balance plus the stock-flow adjustment	108	0.000	2000
Source agreement	PORDATA bridge against the CFP workbook, General Government	31	0.498	1995
Source agreement	PORDATA bridge against the CFP workbook, Central Government	26	66.811	2002
Source agreement	PORDATA bridge against the CFP workbook, Regional and Local	26	0.492	2002
Source agreement	PORDATA bridge against the CFP workbook, Social Security Funds	26	0.470	2019
Vintage revision	1995 modern vintage against the historical vintage, four balances	4	72.465	1995

4 Empirical Framework

Every quantity in this section is an accounting transformation of published data. Nothing is estimated except the piecewise means of Section 6, which are reported with their sensitivity.

4.1 Level decomposition

The identity of Equation (1) is applied year by year. Balances are expressed both in million euro and as a share of contemporaneous nominal GDP, the latter for comparability across five decades

of nominal growth.

4.2 Dynamic attribution

Differencing the identity attributes each annual change to the subsectors:

$$\Delta B_t^{GG} = \Delta B_t^C + \Delta B_t^{RL} + \Delta B_t^{SSF}. \quad (2)$$

This separates the *level* of a subsector balance from its *contribution to a movement*, which are distinct and which the level series alone cannot distinguish: a subsector can be in persistent surplus and still account for a deterioration.

Ranking episodes on nominal euro would rank them by how recent they are, since nominal GDP in 2025 is orders of magnitude above 1977. We therefore also report each term scaled by current-year GDP:

$$\frac{\Delta B_t^{GG}}{GDP_t} = \frac{\Delta B_t^C}{GDP_t} + \frac{\Delta B_t^{RL}}{GDP_t} + \frac{\Delta B_t^{SSF}}{GDP_t}. \quad (3)$$

Because the four terms share one denominator, Equation (3) decomposes exactly, and its residual is Equation (2)'s residual rescaled and nothing more. It is emphatically **not** the change in the balance ratio, $\Delta(B_t^{GG}/GDP_t)$, which also moves with the denominator and does not decompose additively. We use the scaled form for ranking and comparison and the level form for magnitudes.

4.3 Revenue and expenditure attribution

For each sector-year with detailed components the balance is an identity, $B_{i,t} = R_{i,t} - E_{i,t}$, and so is its change:

$$\Delta B_{i,t} = \Delta R_{i,t} - \Delta E_{i,t}. \quad (4)$$

Changes are computed only between adjacent years inside a continuous source block, so no change spans the 1995-to-2000 subsector gap. A caution on reading Equation (4): a balance improves when revenue rises faster than expenditure, so a positive expenditure term alongside an improving balance means expenditure grew but grew less. It is not decomposed into policy and macroeconomic components, which would require assumptions this paper does not make.

Equations (2) and (4) are computed on different source families — the canonical balance panel and the detailed account panel. They therefore measure the same annual change slightly differently, and we carry the residual between them as its own column rather than reconciling it away.

4.4 Offset metric

Writing the combined non-Social-Security balance as $B_t^{nonSSF} = B_t^C + B_t^{RL}$, the offset ratio is defined only when the signs make the comparison meaningful, that is when $B_t^{nonSSF} < 0$ and $B_t^{SSF} > 0$:

$$O_t = \frac{B_t^{SSF}}{|B_t^{nonSSF}|}. \quad (5)$$

$O_t = 1$ means the positive Social Security balance is exactly the size of the negative non-Social-Security balance. The ratio compares two recorded numbers. It is left undefined, and missing, in every year where the sign condition fails, and it is not a counterfactual: see Section 2.

4.5 Persistence

Each annual balance is classified by sign. We report, per subsector, the frequency of each sign, the mean and median balance ratio, the longest unbroken run of each sign, and the empirical one-year sign transition frequencies.

Two conventions follow from the 1995 splice. Magnitudes are computed inside each regime, because a mean that averages two vintages with different levels describes neither. Runs are *not* recomputed per regime: a run is a property of the uninterrupted series, and truncating it at a window boundary would report the length of the window rather than the length of the run. Transition frequencies are empirical frequencies over the transitions actually observed, not a fitted Markov model: no standard errors and no stationarity test are claimed, and a state that never occurs has no estimated row.

4.6 Piecewise-constant mean shifts

Fewer than fifty annual observations split across two regimes do not support a flexible change-point model. The specification is deliberately restrictive: a piecewise-constant mean with at most two breaks per regime, a minimum segment length of five years, exact dynamic-programming minimisation of within-segment squared error, and BIC selection, which may return zero breaks. Segment means and break locations are both counted as free parameters, which penalises an extra break twice and makes the model less willing to claim one. Detection runs separately inside 1977–1994 and 1995–2025, so the known splice cannot be selected as an economic break.

Because a single selected date from eighteen or thirty-one observations is not determined by the data, we report it with a BIC margin over the next-best break count and with a sensitivity grid over the two tuning parameters. The results are in Section 6 and Appendix A, and the dates are described as candidates throughout.

5 Results

5.1 Long-run composition

Figure 1 plots the four balance ratios over the full window. The visual impression is of two series that track each other closely and two that stay near zero: General Government and Central Government are nearly indistinguishable for long stretches, while Regional and Local Government and the Social Security Funds move in a narrow band.

That closeness is itself the first result. It says the aggregate balance has been predominantly a Central Government quantity in level terms, which is what makes the recent divergence at the right-hand edge of the figure notable rather than routine.

Figure 2 draws the identity directly, stacking the signed subsector contributions against the aggregate total. Where a tall stack sits under a shallow line, the subsectors moved in opposite directions and offset one another.

5.2 Persistence, and why magnitudes are regime-specific

Table 4 reports sign frequencies and magnitudes inside each statistical regime.

Three patterns are stable across both regimes. The Central Government balance is negative in every observation of the panel, all 49 of them. The Social Security balance is positive in 43 of 49, negative in 6, with a longest unbroken positive run of 24 years. Regional and Local Government is the only subsector that changes sign with any regularity, positive in 18 years, and it is small in both directions.

Magnitudes, by contrast, are strongly regime-dependent, and this is the reason the table is split. The aggregate balance averages -6.43% of GDP over 1977–1994 and -4.04% over 1995–2025.

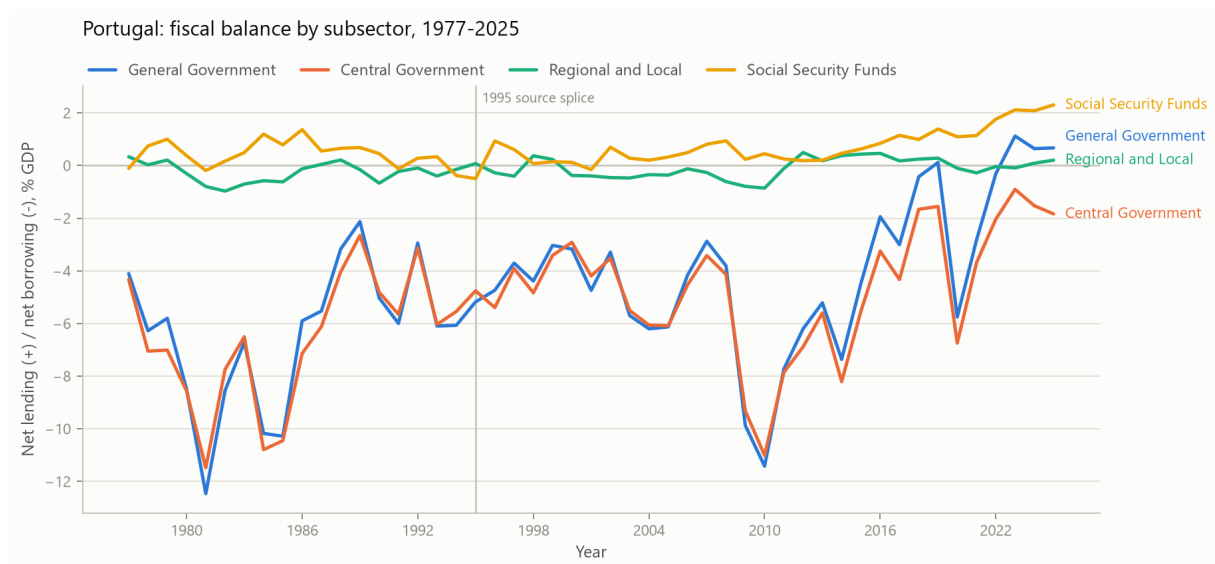


Figure 1: Net lending (+) and net borrowing (-) by subsector, 1977–2025, as a percentage of nominal GDP. The vertical rule marks the 1995 source splice; the series either side of it come from different source families and are not chained. Sources: Banco de Portugal / INE long series and INE via PORDATA.

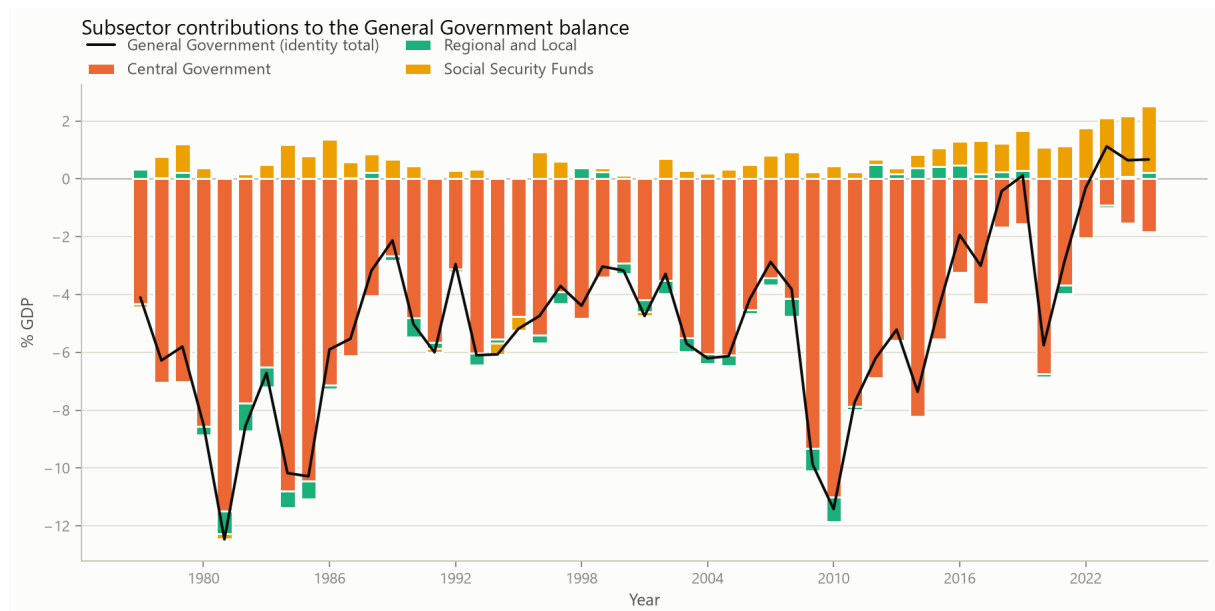


Figure 2: The identity of Equation (1) drawn: signed subsector contributions stacked against the General Government total, as a percentage of GDP. Positive and negative contributions stack away from zero independently, so the visible column height equals the arithmetic sum of the layers.

Table 4: Subsector composition of the general-government balance, by statistical regime. Magnitudes are reported inside each regime because the two differ enough that a pooled mean describes neither.

Regime	Subsector	N	Positive	Negative	Mean (% GDP)	Median (% GDP)
1977–1994 historical	General Government	18	0	18	-6.43	-6.03
1995–2025 modern	General Government	31	4	27	-4.04	-4.17
1977–1994 historical	Central Government	18	0	18	-6.61	-6.31
1995–2025 modern	Central Government	31	0	31	-4.67	-4.33
1977–1994 historical	Regional and Local	18	5	13	-0.27	-0.19
1995–2025 modern	Regional and Local	31	13	18	-0.09	-0.10
1977–1994 historical	Social Security Funds	18	14	4	0.46	0.47
1995–2025 modern	Social Security Funds	31	29	2	0.72	0.61

A pooled mean of -4.91% falls between the two and corresponds to no period in the data. Central Government moves from -6.61% to -4.67% and the Social Security Funds from 0.46% to 0.72%.

Sign counts survive pooling far better than magnitudes, because a sign does not depend on the level convention of a vintage. That asymmetry is why we pool the counts in the discussion above and never pool the means.

5.3 The years with a positive aggregate balance

The aggregate balance is positive in 4 of 49 years: 2019, 2023, 2024 and 2025. Table 5 gives the composition of each.

Table 5: Every year in which the aggregate balance is positive, with its subsector composition and the offset ratio.

Year	GG (M EUR)	Central (M EUR)	Regional/local (M EUR)	SSF (M EUR)	Non-SSF (M EUR)	Offset ratio
2019	249	-3,331	604	2,975	-2,727	1.091
2023	3,030	-2,447	-243	5,720	-2,690	2.126
2024	1,863	-4,424	256	6,031	-4,168	1.447
2025	2,059	-5,636	630	7,065	-5,006	1.411

The composition is the same in all 4. The combined non-Social-Security balance is negative, the Social Security balance is positive, and the latter exceeds the former in absolute size — which is precisely the condition under which the offset ratio of Equation (5) is defined and exceeds one. In 2025 the identity reads

$$2,059 = -5,636 + 630 + 7,065 \text{ M EUR},$$

with a combined non-Social-Security balance of -5,006 million euro and an offset ratio of 1.411. In ratio terms the aggregate balance was 0.67% of GDP against a Social Security balance of 2.30%.

Figure 3 plots the ratio over the years in which it is defined.

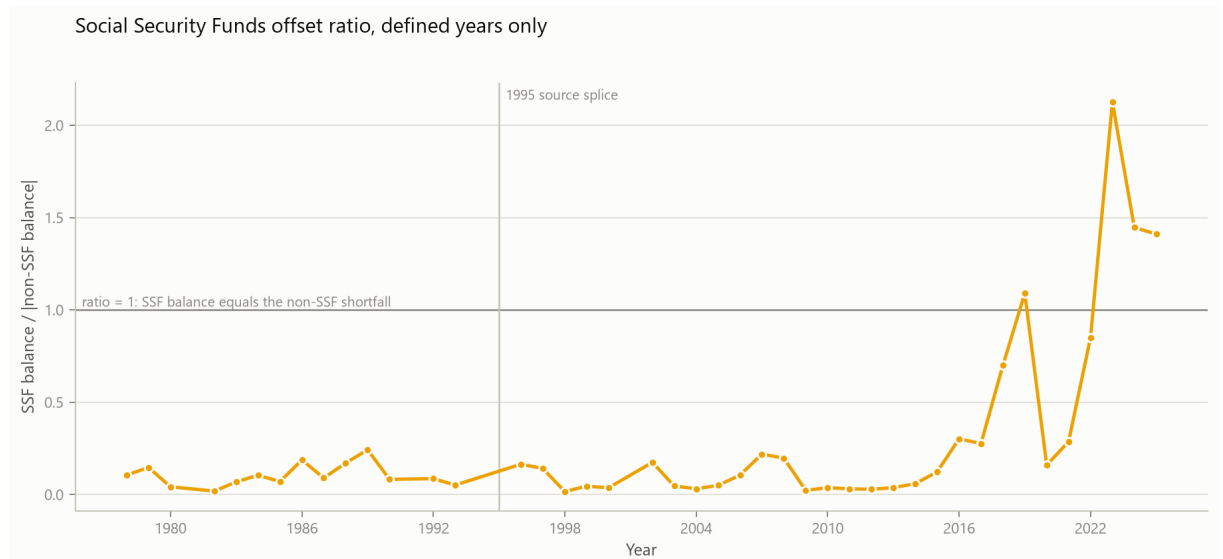


Figure 3: The offset ratio of Equation (5), for the years in which its sign condition holds. The reference line at one marks the level at which the Social Security balance equals the non-Social-Security shortfall. The ratio relates two recorded quantities and is not a counterfactual.

We restate the caution of Section 2 because this is the result most exposed to over-reading. That the ratio exceeds one in 2025 is a statement about the relative size of two published numbers.

It is not a statement that the aggregate balance would have been negative under a different institutional arrangement, and the identity contains nothing that could support such a statement.

5.4 Annual movements

Figures 4 and 5 attribute each annual change to the subsectors, on the GDP scaling of Equation (3). The two windows are drawn as separate panels and neither crosses 1995: a single panel spanning the splice would place a vintage revision among the economic movements and give it the same visual weight.

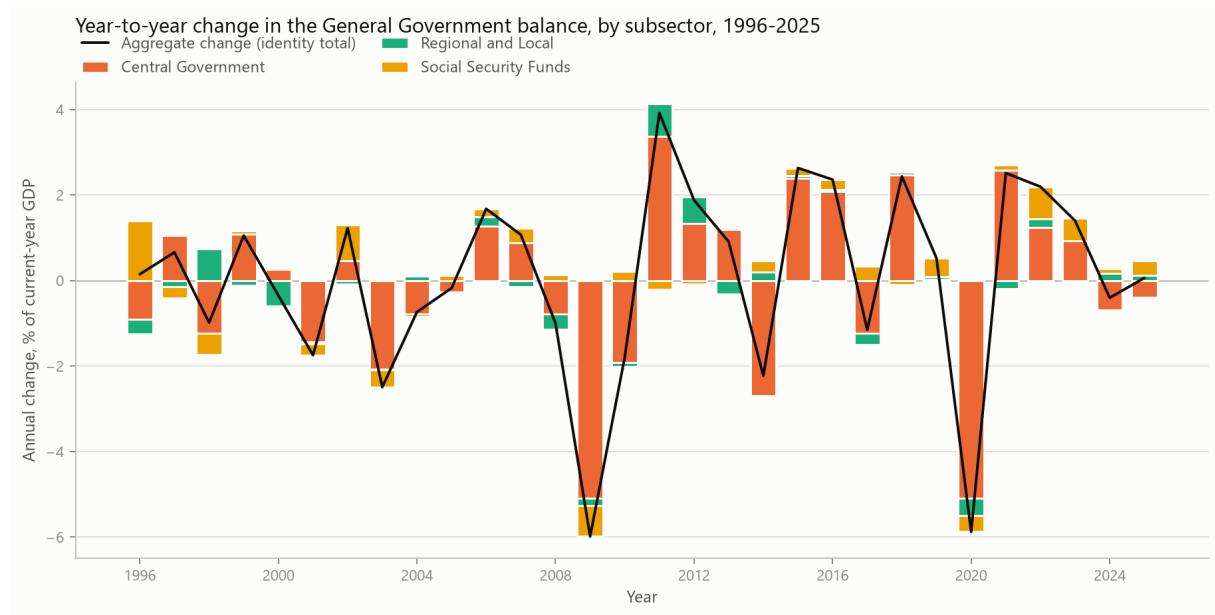


Figure 4: Annual change in the General Government balance attributed to subsectors, 1996–2025, as a percentage of current-year GDP. The black line is the identity total.

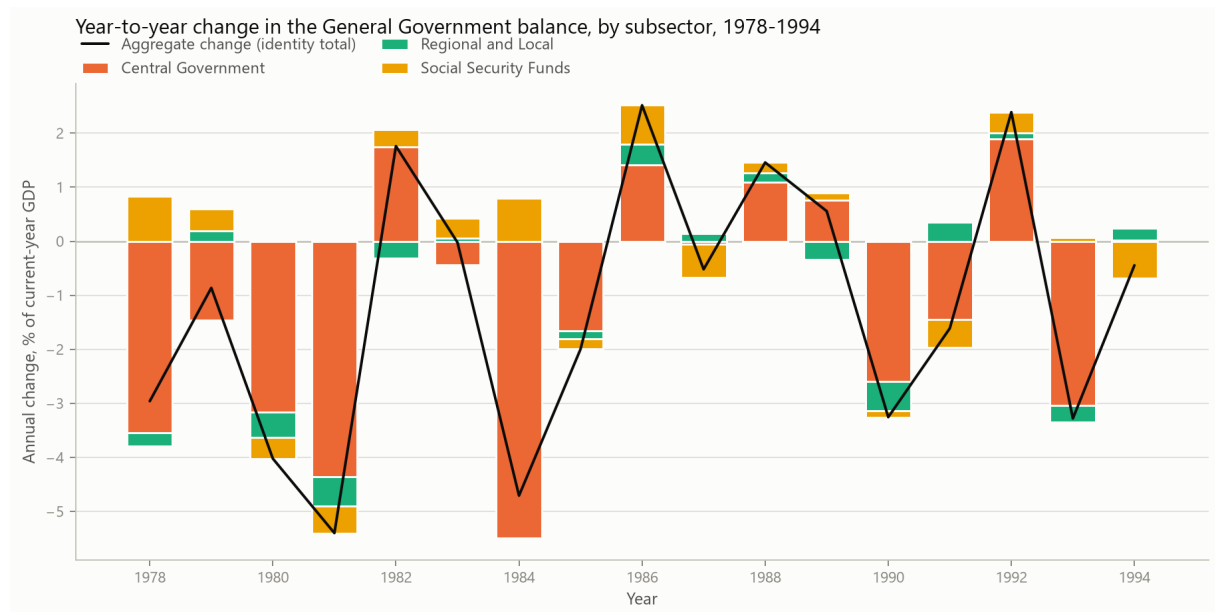


Figure 5: The same decomposition for 1978–1994, on the same scaling. The windows are plotted separately because no change is computed through the 1995 splice.

Table 6 ranks the 5 largest improvements and the 5 largest deteriorations on that scaling, and attributes each to a subsector and then to the revenue and expenditure movements behind it.

Table 6: The largest annual improvements and deteriorations, ranked on the change scaled by current-year GDP, with the subsector and the revenue or expenditure movements that composed each one.

Direction	Year	Change (% GDP)	Central (M EUR)	Regional/local (M EUR)	SSF (M EUR)	From revenue (M EUR)	From expenditure (M EUR)
Improvement	2011	3.92	5,937		1,342	-366	1,796
Improvement	2015	2.63	4,283		121	321	1,854
Improvement	2021	2.52	5,570		-394	270	9,188
Improvement	1986	2.52	355		97	181	2,092
Improvement	2018	2.44	5,053		151	-208	4,794
Deterioration	2009	-5.98	-8,939		-289	-1,268	-3,802
Deterioration	2020	-5.88	-10,224		-811	-777	-4,176
Deterioration	1981	-5.40	-385		-48	-45	573
Deterioration	1984	-4.71	-922		-1	133	943
Deterioration	1980	-4.02	-233		-34	-29	523

1995 is excluded: the 1994-to-1995 change straddles the vintage splice in both panels and mixes a statistical revision with an economic movement.

The scaling changes what the ranking shows, which is the point of using it. 4 of the ten episodes fall in the historical regime — episodes that a nominal ranking places nowhere near the top, because the euro amounts involved are small relative to modern GDP. The largest single improvement is 2011 at 3.92 percentage points of GDP and the largest deterioration is 2009 at -5.98.

Central Government accounts for the largest share of every one of the ten episodes. Given that it also carries the level of the aggregate balance (Section 5), this is less surprising than it first appears, and it is worth stating what it does not mean: the subsector accounting arithmetically for a movement has not been shown to have produced it.

5.5 Social Security mechanisms

We now turn to the second research question. Two distinct layers have to be kept apart, and the value of the exercise lies in the separation.

The national-accounts layer. The ESA 2010 Social Security Funds sector is the entity that appears in Equation (1). In 2025 social contributions were 67.07% of its total revenue. Figure 6 plots that share over the panel.

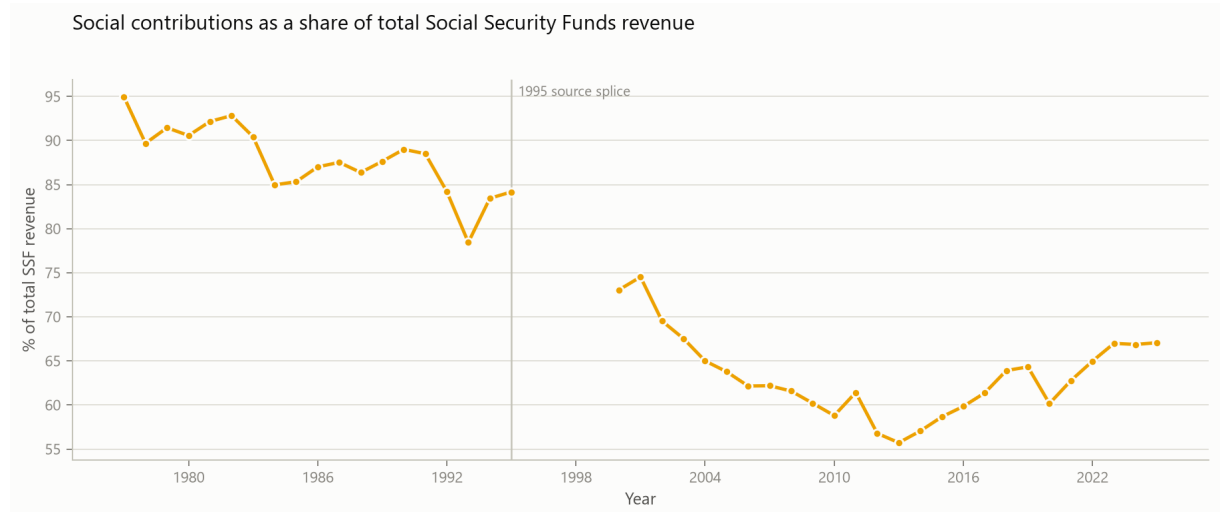


Figure 6: Social contributions as a share of total Social Security Funds revenue. The line breaks at 1996–1999, where subsector components do not exist in either source family.

The budget-execution layer. The CFP publishes the system decomposed into the Previdential system, the Social Protection of Citizenship system and special regimes. These are

budget-execution aggregates on a different accounting boundary. Figure 7 shows them.

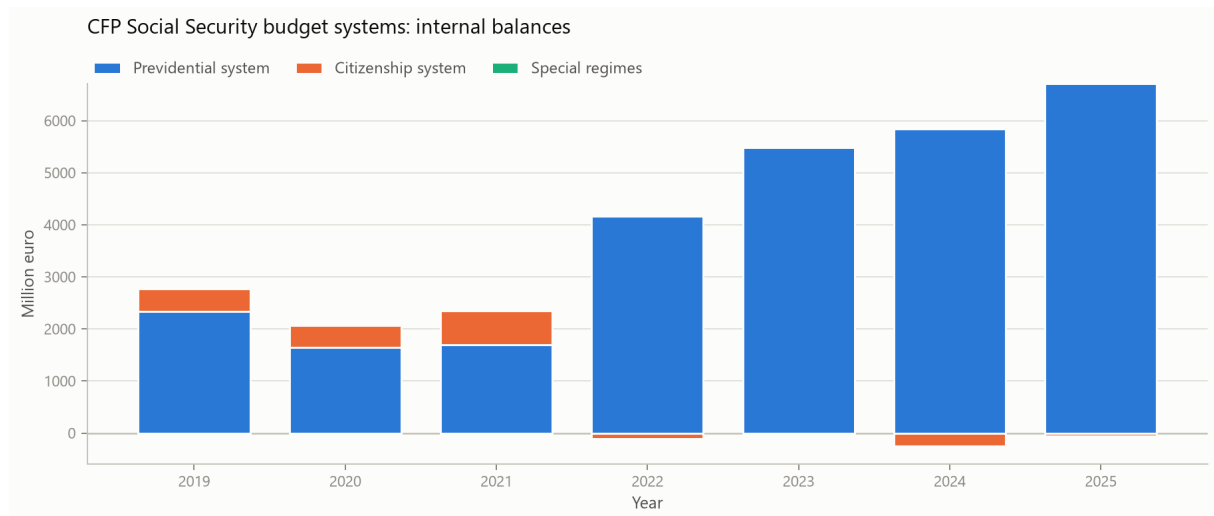


Figure 7: CFP internal Social Security balances by system, in million euro. The movement over the published window is concentrated in the Previdential system.

The movement is concentrated in one system. The Previdential balance goes from 2,333 million euro in 2019 to 6,712 million in 2025, while the Citizenship balance stays small in both directions, standing at -55 million in 2025. Since contributions are the dominant revenue item of the Previdential system, and since the contribution share of national-accounts revenue rose over the same years, the two layers are telling a consistent story about where the movement sits. Neither identifies why it happened, and we do not speculate: a contributory balance moves with employment, wages, contribution rates, entitlement rules and demographics simultaneously, and separating them needs a model this paper does not build.

Why the two layers must not be interchanged. This is a methodological result, and it is the one we would most want a reader to carry away. Table 7 places the two side by side.

Table 7: Two accounting boundaries for Social Security, reported side by side. The columns are never added, netted or reconciled.

Year	ESA 2010 (M EUR)	Budget systems (M EUR)	Previdential (M EUR)	Citizenship (M EUR)	Difference (M EUR)
2019	2,975	2,776	2,333	443	199
2020	2,198	2,072	1,649	423	126
2021	2,468	2,348	1,695	653	120
2022	4,299	4,062	4,172	-110	237
2023	5,720	5,486	5,485	1	234
2024	6,031	5,595	5,840	-245	436
2025	7,065	6,657	6,712	-55	408

In 2025 the ESA 2010 balance was 7,065 million euro while the three budget systems summed to 6,657 million, a difference of 408 million. Across the 7 overlapping years the difference ranges from 120 to 436 million euro. It is never zero.

Figure 8 makes the scale problem visible: on a shared axis the two series are nearly identical, and the difference is only legible on an axis of its own.

The consequence is practical. A Social Security figure quoted from the budget documents is not the figure that enters the national-accounts identity, and substituting one for the other introduces an error of a few hundred million euro without any signal that it has happened. We therefore report both and add neither.

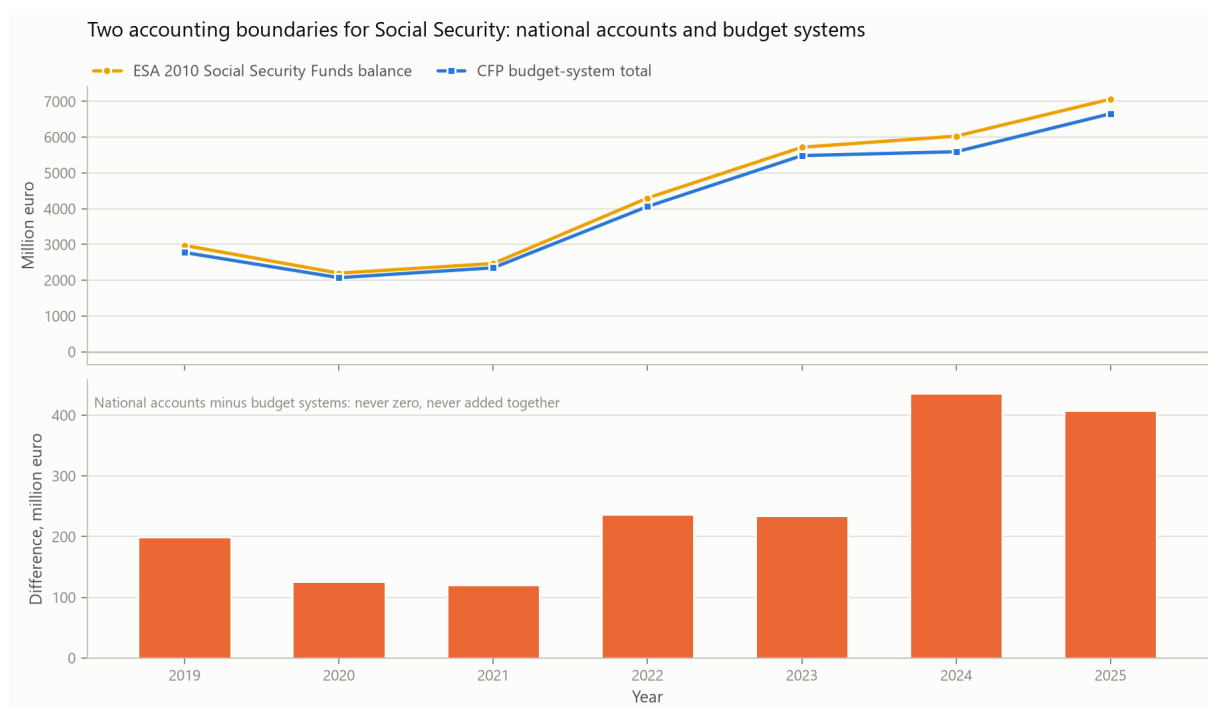


Figure 8: The two Social Security accounting boundaries. Upper panel: both balances on a shared axis, where they are almost indistinguishable. Lower panel: the difference on its own axis. The two are never added, netted or reconciled here.

5.6 Primary balance: the headline sign is not the underlying sign

The Central Government balance is negative in all 49 years of the panel. Read alone, that invites a strong conclusion: that the subsector runs an underlying deficit throughout. The detailed accounts do not support it.

The primary balance is reconstructed as $PB_{i,t} = B_{i,t} + I_{i,t}$, where I is interest expenditure, and checked against the published primary-balance row. Table 8 contrasts the two sign frequencies.

Table 8: Headline against primary balance sign frequencies, over the sector-years for which interest expenditure is published.

Subsector	N	Headline < 0	Primary > 0	Mean interest (% GDP)	Peak interest (% GDP)
General Government	49	45	22	4.16	7.89
Central Government	45	45	15	4.17	7.73
Regional and Local	45	29	20	0.13	0.62
Social Security Funds	45	6	39	0.00	0.04

Over the 45 years for which interest expenditure is published, the Central Government headline balance is negative in 45 — all of them — while the primary balance is positive in 15: 1986, 1987, 1988, 1989, 1990, 1991, 1992, 1994, 1995, 2016, 2018, 2019, 2023, 2024 and 2025. Those years are not confined to the recent period; they run from 1986 to 2025. Interest averaged 4.17% of GDP over the panel and peaked at 7.73%, and that is the arithmetic separating the two series. Figure 9 shows the primary balance crossing zero repeatedly while the headline balance never does.

Both statements are descriptive and they do not conflict. The correct summary is that the Central Government B.9 is negative throughout the panel *and* that its primary balance is positive in a non-trivial minority of the observed years.

The symmetric caution applies. A positive primary balance is not a sustainability result:

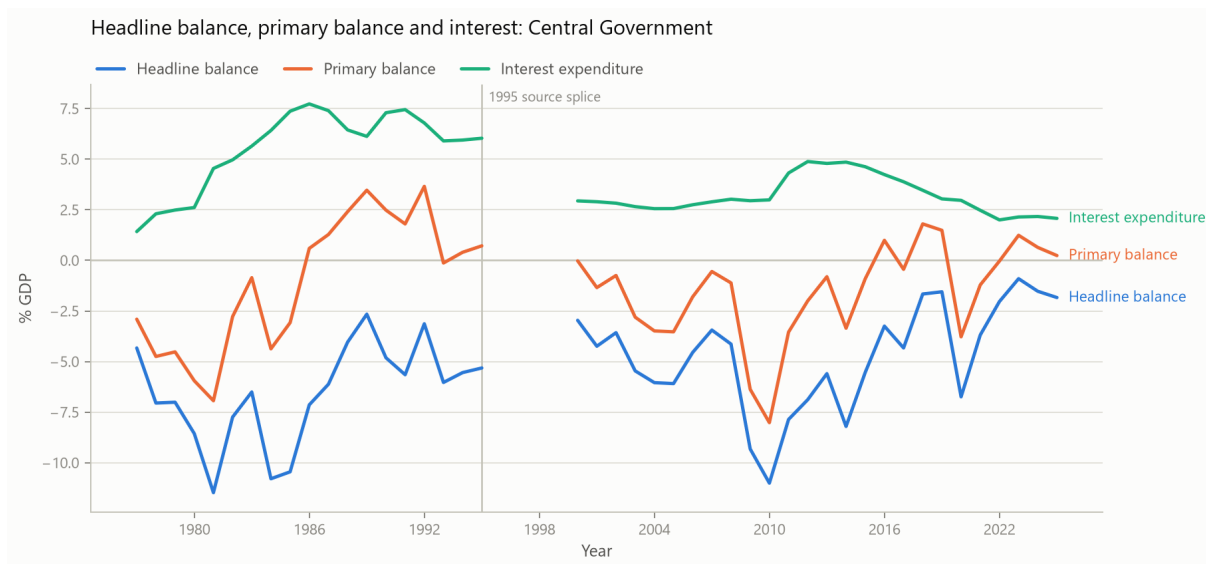


Figure 9: Central Government headline balance, primary balance and interest expenditure, as shares of GDP. The primary balance crosses zero repeatedly; the headline balance does not. The line breaks at 1996–1999.

it excludes interest by construction and therefore says nothing about the debt path, which depends on the interest burden it removes. Neither series, on its own, licenses a conclusion about sustainability.

6 Robustness and Accounting Boundaries

6.1 The 1995 splice

The splice is the single largest threat to the level results, and it is handled by restriction rather than by adjustment. Both 1995 vintages are retained; the revision between them is non-zero for every subsector; no level statistic is averaged across the boundary; and no annual change is computed through it. The regime split of Table 4 is the form in which magnitudes are reported precisely because a pooled magnitude would embed the splice silently.

Sign-based results are far more robust to the splice than magnitudes, since a sign does not depend on the level convention of a vintage. The persistence results of Section 5 therefore carry more weight than the means beside them, and we have kept the two visually separate for that reason.

6.2 How firm are the change-point dates?

The piecewise-constant mean model of Section 4 selects break dates, and a selected date reads as more determined than it is. Two guards are reported.

The first is a BIC margin over the next-best break count, which says how decisively the selection beat its alternatives. The second is a sensitivity grid: detection re-run over 12 combinations of the two tuning parameters, neither of which is estimated from the data. A date that survives only one combination is a property of that choice rather than of the series.

Table 9 reports the grid summary, and Appendix A the full detail.

The grid is not a formality: it changes what may be claimed. In 6 of the 8 series the preferred specification returns the modal set of dates. In the remaining series it does not, and for those the selected dates follow the tuning choice. The least stable case is Regional and Local in the modern regime, where the modal dates hold in only 33% of the grid across 6 distinct date sets.

Table 9: Sensitivity of the detected mean shifts to the two tuning choices, over twelve specifications per series.

Regime	Subsector	Modal breaks	Modal dates	Share at modal dates	Distinct date sets
1977–1994 historical	General Government	1	1986	0.83	2
1995–2025 modern	General Government	2	2009, 2015	0.42	4
1977–1994 historical	Central Government	1	1987	1.00	1
1995–2025 modern	Central Government	2	2009, 2016	0.67	2
1977–1994 historical	Regional and Local	0	–	0.83	2
1995–2025 modern	Regional and Local	2	2012	0.33	6
1977–1994 historical	Social Security Funds	0	–	1.00	1
1995–2025 modern	Social Security Funds	1	2017	0.67	3

Accordingly we describe every date as a candidate rather than a finding, and no break year should be quoted from this paper without its share. We also attach no historical cause to any of them, and note that a five-year minimum segment means a shift in the last four years of the sample cannot be detected at all.

6.3 Intergovernmental transfers and the location of a balance

Because the subsectors are consolidated, a transfer between them changes the components and not the aggregate. This makes the *location* of a balance within general government partly a function of the transfer convention, which matters for interpreting composition results.

The historical source tables identify current and capital transfers between public administrations, which supports a purely mechanical sensitivity for 1977–1995:

$$B_{i,t}^{sens} = B_{i,t} - (T_{i,t}^{received} - T_{i,t}^{paid}).$$

We compute it and deliberately decline to label it an underlying or true balance, and we never use it to restate B.9. The reason is substantive rather than cautious: a transfer normally finances an expenditure responsibility assigned to the recipient, so removing the transfer while leaving the responsibility in place does not describe a coherent alternative arrangement. Which transfer discharges which statutory obligation is a legal question, not an accounting one. What the sensitivity quantifies is how much the recorded location of a balance depends on the convention — and it is reported for that purpose only.

The same reasoning is why no Social Security balance in this paper has State transfers netted out of it.

6.4 What the balance does not determine

Two limits on the reach of these results are worth stating explicitly, because both concern inferences a reader may reasonably be tempted to draw.

The balance does not determine the debt path. The change in debt equals minus the balance plus a stock-flow adjustment that absorbs financial-asset transactions, valuation effects and timing differences, and in several years in these data the adjustment is the larger of the two terms. Whether such residuals reflect accounting responses to fiscal rules (von Hagen and Wolff, 2006) or are smaller and less systematic than that literature suggested (Seiferling, 2013) is a live question on which we take no position. Its relevance here is narrower: a composition result about balances is not a result about debt.

The balance does not identify a mechanism. Nothing in this paper separates cyclical from discretionary movements, and no cyclical adjustment is attempted. A specification relating balances to nominal GDP growth is available in the repository and is reported there in an appendix, where its weaknesses are set out; it is deliberately not part of this paper’s evidence,

because its dependent variable and its regressor share a construction through nominal GDP and the fits are very low. A better-founded mechanism for the Social Security side would model the contribution base directly, regressing the change in contributions on the change in the aggregate wage bill $W_t = N_t \bar{w}_t$ rather than on aggregate nominal growth. We flag it as the natural next step and do not implement it here.

7 Conclusion

Portugal’s general-government balance, decomposed across institutional subsectors over 49 years, has a composition that is asymmetric and remarkably stable. The Central Government balance is negative in every one of the 49 observations. The Social Security balance is positive in 43, with a longest unbroken run of 24 years. Regional and Local Government is the only subsector that changes sign with any regularity, and it is small in both directions. The aggregate is positive in 4 years, and in each of them the combined non-Social-Security balance is negative while the Social Security balance exceeds it in absolute size.

Two results cut against readings that the aggregate series invites.

The first concerns Central Government. Its balance is negative throughout the panel, but its primary balance is positive in 15 of the 45 years for which detailed accounts exist, from 1986 to 2025. A permanent headline deficit is not a permanent underlying one, and interest — averaging 4.17% of GDP and peaking at 7.73% — is the arithmetic that separates them.

The second concerns Social Security. The ESA 2010 Social Security Funds sector and the CFP budget-execution systems are different accounting objects whose balances differ in every one of the 7 years they overlap, by between 120 and 436 million euro. Quoting one where the other belongs introduces an error of that order with no signal that it has occurred.

7.1 What this paper does not establish

The decomposition is an accounting identity. It reallocates a recorded quantity and estimates nothing, and three inferences do not follow from it.

It does not license a counterfactual. That a positive Social Security balance exceeds a negative non-Social-Security balance in a given year says those two published numbers stand in a particular ratio. It says nothing about what the aggregate would have been under a different institutional arrangement, because that counterfactual would move contribution rates, entitlements and transfers together and the identity models none of them.

It does not attribute responsibility. A subsector accounting arithmetically for most of a movement has not been shown to have produced it.

It does not speak to sustainability. Neither the headline nor the primary balance determines the debt path, which also depends on stock-flow adjustments that are the larger term in several years of these data.

Two further limits are properties of the data rather than of the method. Magnitudes are regime-specific and cannot be pooled across the 1995 splice; the pooled aggregate mean of -4.91% of GDP describes neither the -6.43% of 1977–1994 nor the -4.04% of 1995–2025. And 2024 and 2025 are provisional at source: they are the years discussed in most detail here and the years most likely to be restated by a later vintage.

7.2 Extensions

Three extensions follow naturally, in ascending order of what they would add.

A wage-bill specification would give the Social Security results an economic mechanism, relating the change in contributions to the change in the aggregate wage bill rather than to aggregate nominal growth.

A decomposition of revenue into its components would sharpen the episode attribution of Table 6, which currently uses totals and therefore cannot see composition shifts within revenue.

Most valuable would be a European benchmark. ESA 2010 requires the same subsector breakdown from every member state, and Eurostat and the ECB publish it (Eurostat, 2025; European Central Bank, 2025). Defining, for each country and year, $N_{c,t} = B_{c,t}^C + B_{c,t}^{RL}$ and, where the sign condition holds, $O_{c,t} = B_{c,t}^{SSF} / |N_{c,t}|$ would place Portugal in a distribution and answer the question this paper cannot: whether the composition documented here is unusual or ordinary. As a single-country longitudinal study, nothing above can distinguish a Portuguese peculiarity from a general feature of European fiscal accounting.

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A Change-Point Sensitivity in Full

Section 6 summarises the sensitivity of the piecewise-constant mean shifts. This appendix records the specification and the full grid, so the summary can be checked rather than taken.

A.1 Specification

For a balance ratio $y_1, \dots, y_n$ inside one statistical regime, the model partitions the index range into $k + 1$ contiguous segments and fits a constant to each. The partition minimises total within-segment squared error,

$$\min_{0=b_0 < b_1 < \dots < b_k < b_{k+1}=n} \sum_{j=0}^k \sum_{t=b_j+1}^{b_{j+1}} \left(y_t - \bar{y}_{[b_j, b_{j+1}]} \right)^2,$$

solved exactly by dynamic programming rather than by a greedy or binary-segmentation heuristic. Selection over k uses

$$\text{BIC}(k) = n \log \left(\frac{\text{RSS}(k)}{n} \right) + (2k + 1) \log n,$$

where the parameter count $2k + 1$ charges for $k + 1$ segment means and k break locations. Charging for the locations is the conservative choice: it penalises an additional break twice and makes the model less willing to claim one. Zero breaks is always admissible.

Two tuning parameters are not estimated: the minimum segment length and the maximum number of breaks. The baseline sets them to five years and two breaks per regime.

A.2 The grid

Detection is re-run over the product of

$$\text{minimum segment} \in \{4, 5, 6, 7\} \quad \text{and} \quad \text{maximum breaks} \in \{1, 2, 3\},$$

giving 12 specifications for each of the 8 regime-subsector series. Three quantities are persisted for each series: the modal number of breaks and the share of the grid selecting it, the modal set of dates and the share of the grid returning exactly that set, and the number of distinct date sets the grid produced.

The last of these is the most informative single number. A series whose grid returns one date set is telling us something about the series; a series whose grid returns six is telling us about the tuning parameters.

A.3 Reading the result

6 of 8 series have a preferred specification that agrees with the modal grid outcome. Where the two disagree, the modal column is the more reliable summary and the preferred date should not be quoted alone. The least stable series is Regional and Local in the modern regime, at 33% modal agreement across 6 distinct date sets.

Two further limits apply regardless of the grid. A minimum segment length of five years means a shift within the last four years of the sample is undetectable by construction. And selecting a mean shift does not imply the series is piecewise-constant; it is the best fit inside a restricted model class, chosen because the sample is too short to support a richer one.

The full per-specification grid, the BIC ladder over admissible break counts and the stability summary are persisted as `outputs/tables/structural_break_sensitivity.csv`, `outputs/tables/structural_break_bic_ladder.csv` and `outputs/tables/structural_break_stability.csv`.

B Data and Code Availability

B.1 What is archived

The accompanying repository retains the raw source workbooks as downloaded, a SHA-256 digest of each, the source-specific intermediate extractions, the canonical processed panels, every calculated analysis table, the figures, and the executed notebooks with their outputs stored in place. No stage of the chain from a published workbook to a number in this paper is missing.

B.2 How this paper relates to the technical report

The repository contains two documents and they have different jobs.

The technical report is generated end to end: its prose is a thin restatement of the persisted artefacts, it carries every analysis the pipeline computes, and no part of it is authored by hand. It is the place to look for material this paper omits — the fixed-capital-formation diagnostic, the debt and stock-flow reconciliation, the full transition matrices, the intergovernmental-transfer sensitivity and the descriptive macroeconomic co-movement.

This paper is authored. Its argument, its structure and its prose are written, and it carries only the evidence that argument needs.

What the two share is the arithmetic. Every table in this paper is written by the pipeline from a persisted artefact, and every number in its prose is a LaTeX macro generated from one. A sentence here cannot hold a value the data no longer support: if an artefact changes, the macro changes with it, and a macro that ceases to exist fails the build rather than rendering silently as nothing. The authored text lives in `paper/sections/`; the generated inputs live in `paper/generated/` and are never edited by hand.

B.3 Reproduction

```
poetry install
make all          # pipeline, notebooks, tests
make paper       # this document
```

The pipeline is deterministic: the same bundled inputs produce the same outputs, and the outputs are committed, so a reader can check a number without running anything.

B.4 Known limitations of the archive

The bundled sources are a single vintage. 2024 and 2025 are provisional at source and a later vintage will restate them; reproducing this paper against newer files will therefore reproduce the method but not necessarily every figure. The vintage of each retained file is recorded in Table 1 for that reason.

The 4 missing years of subsector detailed accounts, 1996–1999, are absent from both source families. They are not imputed here and cannot be recovered from the bundled data.